

Problem 6.4

You want to triple your money in 25 years. What is the annual compound interest rate necessary to achieve this?

We use the compound interest formula:

$$A = P(1 + i)^t.$$

Since tripling the money means $A = 3P$, we have

$$3P = P(1 + i)^{25}.$$

Dividing both sides by P ,

$$3 = (1 + i)^{25}.$$

Taking the 25th root of both sides,

$$3^{1/25} = 1 + i.$$

Thus,

$$i = 3^{1/25} - 1.$$

Therefore,

$$i \approx 0.0449.$$

So, the annual compound interest rate needed is

$$\boxed{4.49\%}.$$